

Semester - 1
Subject Code: HM100204
Max. Marks: 40

Session 2025-26
Subject Name: Professional Communication in English

Class Test- 2

Min. Marks: 14

Course Outcomes (CO):
On successful completion of the course, the student will be able to:

CO 1: The students will be able to demonstrate a better understanding of the communication process by identifying, explaining, and applying current communication theories as they relate to various contexts and learn to use standard formats, techniques and principles to gain credibility in business setting.

CO 2: The students will be able to listen, read, comprehend and synthesize information from different sources and respond appropriately.

CO 3: The students will be able to demonstrate different modes of expression in written communication and develop the fundamental skills and techniques in drafting effective business documents.

CO 4: The students will be able to demonstrate the various oral communication situations to elicit desired results.

CO 5: The students will be able to demonstrate the ability to identify and solve common grammar errors to develop grammatical accuracy.

Note: Section 1 is compulsory and no internal choice is given.

Section 2 is compulsory and with internal choices in the questions given.

Section - I				
Q. No.	Questions	Marks	CO	BL
1a)	What do you mean by ABC principal of Precise writing?	1	CO1, CO2 & CO4	L1, L2 & L3
b)	Define Inventory report.	1	CO1, CO2 & CO4	L1, L2 & L3
c)	Fill in the blank An enquiry made in response to the seller's advertisement & publicity is called _____ enquiry.	1	CO1, CO2 & CO4	L1, L2 & L3
d)	Tick the correct one: What is the purpose of a visual aid in a presentation? i) To distract the audience ii) To replace the speaker iii) To support and clarify key ideas iv) To fill empty space	1	CO1, CO2 & CO4	L1, L2 & L3
2a)	What do you mean by agenda & minutes of the meeting?	4	CO1, CO3 & CO4	L1 & L2
b)	List the differences between tender & Quotation.	4	CO2 & CO4	L1, L2 & L3
c)	Punctuate the following passage. Rita was excited for the school trip she had been waiting for it all year dont forget your lunch box her mother reminded her rita smiled and said I won't	4	CO1 & CO5	L2, L4 & L5

Section -- II

1a) Email is an essential tool for communication. Explain. Or Draft a job application along with resume for the post of Junior engineer in a construction firm. Or What guidelines should be followed by an individual for taking an effective part in group discussion?	8	CO1 & CO 4	L2, L4 & L5	9.2.1 9.2.3 9.2.4
b) Discuss how a chairman ensures smooth & productive meetings. Or You are the manager of Bright Computers Ltd. Draft a quotation letter quoting prices for computers, Printers & Accessories	8	CO2 & CO 4	L1, L2 & L3	9.2.1
c) Do as Directed i) Fill in the blanks with present perfect or past simple form of verbs given in brackets. Mr. Seth ____ several books. He ____ his tenth book last week only. (write, finish) ii) Fill in the blanks with Simple past or past continuous form of verbs given in the brackets: We had covered quite a long distance when we suddenly ____ that we ____ in the opposite direction. (realize, travel) iii) Rewrite the sentences in to indirect speech. a) "Don't post the letter without showing it to me", my father said to me. b) She says, "I am happy to be here." iv) Put the following sentence in to passive voice. a) A special committee will conduct an enquiry. b) The principal asked the student to explain the cause of the failure. Or What practical guidelines can be offered to help interviewees in facing the interview board?	2 2 2 2 8	CO2 & CO3 CO2 & CO3	L1 & L2 L1 & L2	10.1.1 10.1.3 10.1.1 10.1.3

CT, Nov-Dec, 2025

Course & Sem: B.Tech, I Sem
Subject Code: CS100203
Max Marks: 40

Subject Name: PPS
Minimum Pass Marks: 14

Branch: CSE
Section – E2 & H2
Time: 2Hr

Section-I: Multiple Choice Questions

All questions are compulsory. (4 × 1 = 4 Marks)

Q1. Which of the following correctly declares an integer array?

- A. int arr();
- B. int arr[5];
- C. int arr = 5;
- D. int arr{5};

Q2. What does the expression $*(p + 2)$ do, if p is a pointer to an array?

- A. Accesses the first element
- B. Accesses the second element
- C. Accesses the third element
- D. Gives an error

Q3. Which operator is used to access the value stored at a pointer?

- A. &
- B. *
- C. #
- D. %

Q4. What will be output

```
int i = 1;  
do {  
    printf("%d ", i);  
    i++;  
} while(i < 1);
```

- A) No output
- B) 1
- C) 1 2
- D) Infinite

Section-II: Short Answer Questions

Solve any three. ($3 \times 4 = 12$ Marks)

Q1. Draw the given pattern using for loop

```
1
01
101
0101
```

Q2. What is recursive and write a C program using recursive function for finding factorial of a number.

Or

Write a C program using recursive function for finding sum of n natural numbers.

Q3. Explain array of structure with an example.

Or

Write a programme to sort the elements of an array using Bubble Sort algorithm.

Section-III: Descriptive/Numerical Questions

All questions are compulsory. ($3 \times 8 = 24$ Marks)

Q1. What is function? Write difference between call by value and call by reference method.

Or

Write a C Program to rotate a matrix by 90 degree (clockwise).

Q.2 Define string. How string is declared and initialized? Explain string input/output functions with an example.

Or

Write a C Programme to determine whether the string is palindrome or not.

Q.3 Difference between exit controlled loop and entry controlled loop with example.

Or

Write a C Program to generate all Armstrong No. between 1 to 1000.



Roll No:

Total Printed Pages: 2

AM100101

CT- II (December 25)

Subject- Applied Maths-I

B. TECH – I SEM (Q1 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hours

Note:-

- 1) All questions are compulsory.

Section -1

Q. (1) Objective Questions:

(1x4=4)

- (a) Find the Jacobian of the following transformation:

$$u = ax+by, v = cx+dy$$

- (b) If $r = f(\theta)$ then length =

(c) $\int_0^{\pi} \cos^6 \theta = \dots \dots \dots$

- (d) Find the Complementary Function of:

$$(D^2+2D+1)y = 0.$$

Q. (2) Short Answer Type Questions:

(4x3=12)

1. Find the maximum and minimum value of:

$$u = x^3 - 4xy + 2y^2.$$

2. Find the reduction formula of: $\int \tan^n x dx$.

3. Find the whole area of the cardioid, $r = a(1+\cos \theta)$.



Section -2

Q. (3) Descriptive Answer Type Questions:

(8x3=24)

1. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of the unite sphere $x^2 + y^2 + z^2 = 1$.

Or

Find the volume of a sphere of a radius a .

2. Solve: $(D^2 - 2D + 1)y = xe^x \sin x$.

Or

Prove that:

$$\int_0^{\frac{\pi}{2}} \cos^n x \cos nx dx = \frac{\pi}{2^{n+1}}$$

$$P.T. = \frac{1}{f(p)} \times \dots$$

3. Apply the method of variation of parameters to solve;

$$\frac{d^2y}{dx^2} + 4y = \tan 2x.$$

Or

$$\text{Solve: } (2x + 3)^2 \frac{d^2y}{dx^2} - (2x + 3) \frac{dy}{dx} - 12y = 6x$$



Class Test-II, Session – 2025-2026

Subject – Environmental Chemistry

Subject Code – AC100208

Time – 2 hrs

MM - 40

Section -1

Q. 1 Answer in one word or in one sentence - (1×4=4M)

- i) Name two colouring agents used as food additives.
- ii) Give an example of one-dimension nanomaterial.
- iii) What is the botanical name of Aloe vera?
- iv) Write the structure of Dioxin.

Q.2 Short Answer type Questions (3×4=12M)

- i) Write the sources and harmful effects of Asbestos pollution.
- ii) Differentiate Top-down and Bottom-up approach of synthesis of nanomaterials.
- iii) Write the importance of medicinal and aromatic plants in human health.

Section -2

Q.3 Long Answer type Questions: Answer any one question each from A, B and C (3X8= 24M)

- QA.i)** What are food additives? Explain its classification with examples.
- ii) Explain the sources, effects and treatment of effluents of Textile industry.

OR

- i) Explain the classification of polymers with examples.
- ii) What are polychlorinated biphenyls and how do they impact the environment?

QB. Explain how nanoparticles are used for the removal of water contaminants.

OR

Write a brief note on (Any 2):

- i) Nanoporous membrane for water filtration
- ii) Nanocatalyst for gas purification
- iii) Fullerenes

QC. Write the medicinal importance, climatic and soil requirements, cultural practices and harvesting of Mulethi and Lemon grass plant.

OR

Write a brief note on:

- i) Indian knowledge system in organic farming
- ii) Indian legislation in conservation of medicinal and aromatic plants

Section-A

Q1. Answer the following Objective Question- [1x4=04]

i. Which term describes the study of forces causing motion (e.g., muscle force?)

a) Kinematics

☒ b) Kinetics

Mechanics

☒ c) Biomechanics

d) Ergonomics

ii. Which mechanical property describes a tissue's ability to resist deformation?

a) Viscosity

☒ b) Elasticity

c) Conductivity

d) Compressibility

Course Code: ME100207

iii. What is the fundamental principle of 3D printing?

a) Subtractive manufacturing (removing material)

✓ b) Additive manufacturing (adding layers)

c) Molding and casting

ⓓ Laser cutting

iv. Which battery type is commonly used in modern HEVs?

a) Lead-Acid

b) Nickel-Cadmium

✓ c) Lithium-Ion

d) Zinc-Carbon

Q2. Answer the following questions (Short Answer Type)-[4x3=12]

a) Explain the scope of biomechanics?

b) Explain Terminology used in 3d Printing?

c) What is robot anatomy?

Section-B

Attempt any one questions from each unit

[8x3=24]

Unit-III

Q3. Explain the 9 fundamental principle of biomechanics?

Q4. What is implant biomechanics? How biomechanics helps in human body?

Unit-IV

Q5. Explain in brief steps used in 3d Printing?

Q6. What is Industrial 4.0, and which type 3d printing application used in industrial 4.0?

Unit-V

Q7. What is arduino? Also Explain Fundamental Components of mechatronics?

Q8. Elaborate Parts of HEV's, Also Differentiate between electric and Hybrid Electric Vehicles?
